Replication Requirements and Pre-Registered Protocols for Independent Verification of Suppressed-Fatigue Claims in Hierarchical Metallic Alloys

F. Müller[1,*], E. Fernández-Villa[2], H. L. Wong[1], and O. Iskandarova[3]

[1]*Department of Materials Engineering, Zurich Polytechnic, Zurich, Switzerland*
[2]*Laboratorio de Mecánica Experimental, Universidad Autónoma de Madrid, Madrid, Spain*
[3]*Institute for Dislocation Physics, Novosibirsk State Technical University, Novosibirsk, Russia*
[*]f.mueller@zpoly.ch

Abstract

We propose a pre-registered replication protocol for the claim by Kawabata et al. of suppressed fatigue in hierarchical metallic architectures. The protocol specifies specimen preparation, instrumentation, acceptance criteria, and a decision rule for distinguishing confirmation from disconfirmation, all fixed prior to data collection. We identify the subset of measurements that can be completed within 90 days and the subset that requires up to 24 months of continuous cycling. We argue that pre-registration is particularly important for claims of this magnitude, where the small number of research groups with the necessary equipment creates a risk of publication bias in replication. We invite participating laboratories to adopt or modify the protocol and commit to publishing results regardless of outcome.

1 Motivation

Claims of extraordinary effect magnitude in applied mechanics share a recurring failure mode: early enthusiastic replication attempts that return ambiguous data are disproportionately published if they appear to confirm the original claim, while null results are disproportionately not submitted or not accepted. Over time this produces a biased literature in which the apparent effect size declines as replication expands, and consensus eventually forms around a much smaller or null effect.

To reduce this failure mode in the replication of the recent reports of fracture-immune hierarchical alloys [1, 2], we propose a pre-registered replication protocol. Participating laboratories commit, before data collection, to specific procedures, analysis steps, and publication of results regardless of outcome.

2 Pre-registered protocol

2.1 Specimen preparation

1. Feedstock powder chemistry verified by independent wet-chemical analysis within 0.1 wt.% of the reported Fe-18Cr-10Ni-2Mn-0.3N composition.
2. Powder particle size distribution within the reported range of 15 to 45 μm (0.6 to 1.8 mil) with at least 90% of mass by particle count in this window.
3. LPBF processing with recursive hatch pattern as specified in the original supplementary material.
4. ISFA performed at 780 ± 5 °C (1436 ± 9 °F) for 45 ± 1 min under elastic strain of $0.150 \pm 0.005\%$.
5. Independent metallographic verification of hierarchical microstructure at three scales before any mechanical testing.

2.2 Primary measurements

1. Quasi-static tensile tests at nominal strain rate 10^{-3} s^{-1} on $n \geq 6$ specimens per condition.
2. Fully reversed ($R = -1$) fatigue at stress amplitudes $\sigma_a/\sigma_y = \{0.4, 0.6, 0.8, 1.0, 1.1, 1.2\}$ with $n \geq 3$ specimens per amplitude.
3. Fatigue testing at two frequencies per amplitude: nominal 20 kHz resonant, and conventional 20 Hz servo-hydraulic. The two-frequency protocol is specifically designed to detect the frequency-dependent artifacts identified by several commentators.
4. Continuous monitoring of first-mode resonance frequency with resolution better than $\Delta f/f = 10^{-5}$.
5. Surface temperature monitoring with spatial resolution ≤ 0.5 mm (0.02 in) and temporal resolution ≤ 10 s.

2.3 Secondary (orthogonal) measurements

1. Positron annihilation spectroscopy on interrupted specimens at 10^6, 10^8, 10^{10} cycles.
2. High-resolution synchrotron X-ray diffraction peak broadening on the same specimens.
3. Destructive metallography with TEM inspection on at least one specimen per interruption.

2.4 *Decision rule*

We adopt the following pre-registered decision rule:

- *Confirmation.* At least two independent laboratories observe: (i) no statistically significant change in first-mode resonance frequency at $\sigma_a/\sigma_y = 1.0$ through $N \geq 10^{10}$ cycles; AND (ii) absence of persistent slip band morphology in TEM of cycled specimens; AND (iii) agreement with the original monotonic properties within 10%.
- *Disconfirmation.* At least two independent laboratories observe statistically significant damage indicators at $N < 10^9$ cycles at $\sigma_a/\sigma_y = 1.0$, or failure to reproduce the monotonic properties within 25%.
- *Indeterminate.* Results that do not meet either criterion after 24 months of accumulated testing, including cases of intermediate damage rates, narrow processing window sensitivity, or specimen-to-specimen variability exceeding the pre-registered thresholds.

3 Division of labor

We propose the following division of labor across participating laboratories:

- Three laboratories (one each in Europe, North America, East Asia) conduct full 24-month replication with both 20 Hz and 20 kHz cycling.
- Five laboratories conduct 90-day short-cycle validation (up to $N = 10^{10}$) at 20 kHz only.
- Two laboratories conduct *in-situ* TEM fatigue on a subset of specimens.
- One laboratory performs positron annihilation spectroscopy on interrupted specimens from multiple participating labs.

This division allows rapid preliminary confirmation or disconfirmation within 90 days, with full verification at the 24-month horizon.

4 Commitment to publication

All participating laboratories commit to publishing all results, including null results, indeterminate results, and results that disagree with the pre-registered decision rule. Publication venue is not required to be peer-reviewed; a preprint in the appropriate arXiv category within 60 days of experiment completion satisfies the commitment.

5 Discussion

Pre-registration is an uncommon practice in experimental mechanics, where protocol detail is typically reported retrospectively in the methods section. We argue that the specific circumstances of the present claim justify departure from that norm: the number of laboratories with the necessary resonant fatigue capability is small, the cost per replication attempt is high, and the risk of publication bias accordingly large. We note that pre-registration does not preclude exploratory analysis; it merely separates confirmatory from exploratory claims and requires that the former be specified in advance.

6 Invitation

We invite any laboratory with the required capability to join the replication effort, to propose modifications to the pre-registered protocol before adoption, and to publish results according to the above commitment. A public registry of participating laboratories and current status is maintained at an anonymized URL to be disclosed upon acceptance of this note.

References

[1] Y. Kawabata et al., *arXiv* 2604.11891 (2026).

[2] L. Zhang et al., *arXiv* 2604.12207 (2026).

[3] A. Ito et al., *arXiv* 2604.12884 (2026).

[4] T. Johansson et al., *arXiv* 2604.13015 (2026).

[5] S. Bhattacharya, *arXiv* 2604.14728 (2026).